

GLOBAL AI JOBS ANALYSIS

Project Overview

- The Global AI Jobs Analysis project examines worldwide Artificial Intelligence job market trends using a dataset containing 1,500 AI job records.
- The analysis focuses on AI job distribution, salary trends, required technical skills, experience levels, work modes and geographical demand across different countries.
- Using Microsoft Excel for data preprocessing and Power BI for visualization an interactive dashboard was developed to provide meaningful insights that support recruitment strategies, workforce planning and data-driven business decisions.

Objectives

- To analyze the distribution of AI job opportunities across different countries and identify countries with the highest AI job demand.
- To analyze salary variations across different experience levels throughout the year.
- To understand how AI jobs are distributed across education qualifications and experience levels.
- To identify high-paying AI job roles and compare their salary ranges.
- To analyze the relationship between AI job demand and salary levels.
- To visualize the geographical distribution of AI jobs and average salaries across countries.
- To evaluate the distribution of work modes (Fully Remote, Hybrid, and On-site).
- To identify the most in-demand technical skills required for AI professionals.

Dataset Information

- The Global AI Jobs Dataset contains 1,500 records and 13 features related to AI job opportunities.
- It includes information such as job titles, salaries, experience levels, education, work modes, skills, and countries.
- The dataset shows an average salary of \$194,892 with an overall growth rate of 40.38%, making it useful for analyzing global AI job market trends.

Data Preprocessing

The following preprocessing steps were performed to improve data quality:

- Removed duplicate records.
- Identified and handled missing values.
- Standardized categorical values.
- Corrected data types.
- Verified salary and experience-level consistency.
- Imported the cleaned dataset into Power BI for analysis.

Dashboard Analysis

The Power BI dashboard includes:

- KPI Cards
- Country-wise Job Count
- Average Salary by Country
- Work Mode Distribution
- Top 10 Required Skills
- Education vs Experience Analysis
- Salary Range by AI Job Role
- Demand vs Salary Scatter Plot
- Monthly Salary Trend Analysis

Interactive slicers were added to filter the dashboard by Experience Level.

Key Findings

- The analysis shows that the United States has the highest number of AI job opportunities, with more than 500 job postings, making it the leading country in AI employment among the countries analyzed.

- The average annual salary for AI professionals is \$194,892, with monthly average salaries ranging from approximately \$180,000 to \$206,000. The highest average salary was recorded in September (approximately \$205,000), while the lowest average salary was observed in April and July (approximately \$180,000).
- The work mode analysis indicates a balanced distribution of AI jobs, with 34% Fully Remote, 33% Hybrid, and 33% On-site positions, showing that organizations offer flexible working options almost equally.
- Among all the technical skills analyzed, Python is the most in-demand skill, followed by SQL, Statistics, Research, PyTorch, Cloud Computing, Linux, and Kubernetes, highlighting Python as the dominant programming language in AI job roles.
- The education analysis reveals that candidates with Bachelor's and Master's degrees account for the highest number of AI job opportunities. The experience-level analysis also shows that Mid-level (3–5 years) and Senior (6–9 years) professionals represent the largest share of AI hiring.
- The salary analysis by job role indicates that positions such as AI Engineer, Machine Learning Engineer, AI Security Engineer, and Senior Machine Learning Engineer receive some of the highest salaries, with maximum annual salaries reaching approximately \$300,000.
- These findings demonstrate that the AI job market offers strong career opportunities, competitive salaries, flexible work arrangements, and high demand for professionals with Python and machine learning skills.

Business Insights

- Organizations should prioritize hiring professionals with strong Python, SQL, and Machine Learning skills.
- The USA remains the largest AI employment market and offers the greatest hiring opportunities.
- Equal distribution of remote, hybrid, and on-site jobs suggests that companies should continue supporting flexible work policies.
- Competitive salaries exceeding \$190K indicate strong market demand for experienced AI professionals.
- Educational qualifications combined with practical technical skills significantly improve employment opportunities.

Conclusion

- The project successfully analyzed 1,500 AI job records using Excel and Power BI to understand global AI employment trends.
- The dashboard identified the USA as the leading country for AI job opportunities, while Python emerged as the most demanded technical skill across AI roles.
- The analysis revealed an average AI salary of \$194,892, with salaries ranging from approximately \$180K to \$206K throughout the year, indicating a strong compensation trend in the AI industry.
- Work mode analysis showed an almost equal distribution of Fully Remote (34%), Hybrid (33%), and On-site (33%) jobs, reflecting the growing adoption of flexible work environments.
- Overall, the project demonstrates how interactive Power BI dashboards can transform AI job market data into actionable insights, helping organizations optimize recruitment strategies and enabling job seekers to identify high-demand skills, competitive salaries and promising career opportunities.